

Higher Order Derivatives

For each problem, find the indicated derivative with respect to x .

1) $y = -x^2$ Find $\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = -2$$

2) $f(x) = 4x^3$ Find f''

$$f''(x) = 24x$$

3) $y = -4x$ Find $\frac{d^3y}{dx^3}$

$$\frac{d^3y}{dx^3} = 0$$

4) $f(x) = 5x^4$ Find f'''

$$f'''(x) = 120x$$

5) $y = -5x^4$ Find $\frac{d^4y}{dx^4}$

$$\frac{d^4y}{dx^4} = -120$$

6) $y = 3x^5 - 2x$ Find $\frac{d^3y}{dx^3}$

$$\frac{d^3y}{dx^3} = 180x^2$$

7) $y = -2x^3 - 4x^{-3}$ Find $\frac{d^3y}{dx^3}$

$$\frac{d^3y}{dx^3} = -12 + \frac{240}{x^6}$$

8) $y = -x^2 + 2\sqrt[5]{x^2}$ Find $\frac{d^3y}{dx^3}$

$$\frac{d^3y}{dx^3} = \frac{96}{125x^{\frac{13}{5}}}$$

Critical thinking questions. Find the indicated derivatives with respect to x .

9) $y = 99x^{99}$ Find $\frac{d^{100}y}{dx^{100}}$

10) $f(x) = x^{99}$ Find $f^{(99)}$

99! (Made easy by factorial notation)

The 99th derivative is a constant, so 100th derivative is 0.